

Computer Architecture Lab (Week 2)

1. Add two 8-bit numbers

- ❑ Statement: Add the contents of memory locations 4000H and data AA H and place the result in memory location 4002H.

2. Subtract two 8-bit numbers

- ❑ Statement: Subtract the contents of memory locations 4200H and 4201H and place the result in memory location 4202H.

3. Add contents of two memory locations

- ❑ Statement: Add the contents of memory locations 4000H and 4001H and place the result in the memory locations 4002H and 4003H.
- ❑ Problem:
 - (4000H) = 7FH
 - (4001H) = 89H
 - Result = 7FH + 89H = 108H
 - (4002H) = 08H
 - (4003H) = 01H
- ❑ Extra 2: Why AC flag set?
- ❑ Extra 1: Use the same memory location with the same number and find the way to include instruction LHLD and SHLD.

4. Finding one's complement of a number

- ☐ Statement:
- ☐ Find the 1's complement of the number stored at memory location 4400H
- ☐ Store the complemented number at memory location 4300H.

5. Finding Two's complement of a number

- ☐ Statement: Find the 2's complement of the number stored at memory location 4200H
- ☐ Store the complemented number at memory location 4300H.

6. Subtract contents of two memory locations

- ❑ Subtract the contents of memory location 4002H from location 4000H and 4001H and place the result in the memory locations 4003H (low byte) and 4004H (high byte).
- ❑ Problem:
 - (4000H) = D4H
 - (4001H) = 02H
 - (4002H) = EFH
 - Result = $02D4H - EFH = 01E5H$
 - (4004H) = 01H
 - (4003H) = E5H
- ❑ Extra 1: Why AC flag set?
- ❑ Extra 2: Use the same memory location with the same number and find the way to include instruction LHLD and SHLD.

7. Add Two 16-bits numbers with Carry

- ❑ Statement: Add the 16-bit number in memory locations 4000H and 4001H to the 16-bit number in memory locations 4002H and 4003H
- ❑ The most significant eight bits of the two numbers to be added are in memory locations 4001H and 4003H.
- ❑ Store the result in memory locations 4004H and 4005H with the most significant byte in memory location 4005H.
- ❑ Sample Problem:
 - ❑ (4000H) = A5H
 - ❑ (4001H) = 1CH
 - ❑ (4002H) = 97H
 - ❑ (4003H) = 5AH
 - ❑ Result: 1CA5H + 5A97H = 773CH
 - ❑ (4004H) = 3CH
 - ❑ (4005H) = 77H

8. Subtract two 16-bit numbers with

- ❑ Statement: Subtract the 16-bit number in memory locations 4002H and 4003H from the 16-bit number in memory locations 4000H and 4001H.
- ❑ The most significant eight bits of the two numbers are in memory locations 4001H and 4003H.
- ❑ Store the result in memory locations 4004H and 4005H with the most significant byte in memory location 4005H.
- ❑ Sample problem:
 - ❑ (4000H) = 19H
 - ❑ (4001H) = 6AH
 - ❑ (4002H) = 15H
 - ❑ (4003H) = 5CH
 - ❑ Result = 6A19H - 5C15H = 0E04H
 - ❑ (4004H) = 04H
 - ❑ (4005H) = 0EH

9. Subtract two 16-bit numbers (2)

- ❑ Statement: Subtract the 16-bit number in memory locations 4002H and 4003H from the 16-bit number in memory locations 4000H and 4001H.
- ❑ The most significant eight bits of the two numbers are in memory locations 4001H and 4003H.
- ❑ Store the result in memory locations 4004H and 4005H with the most significant byte in memory location 4005H.
- ❑ Sample problem:
 - ❑ (4000H) = E5H
 - ❑ (4001H) = 30H
 - ❑ (4002H) = EFH
 - ❑ (4003H) = 01H
 - ❑ Result = 30E5H - 01EFH = 2EF6H
 - ❑ (4004H) = F6H
 - ❑ (4005H) = 2EH

Thanks